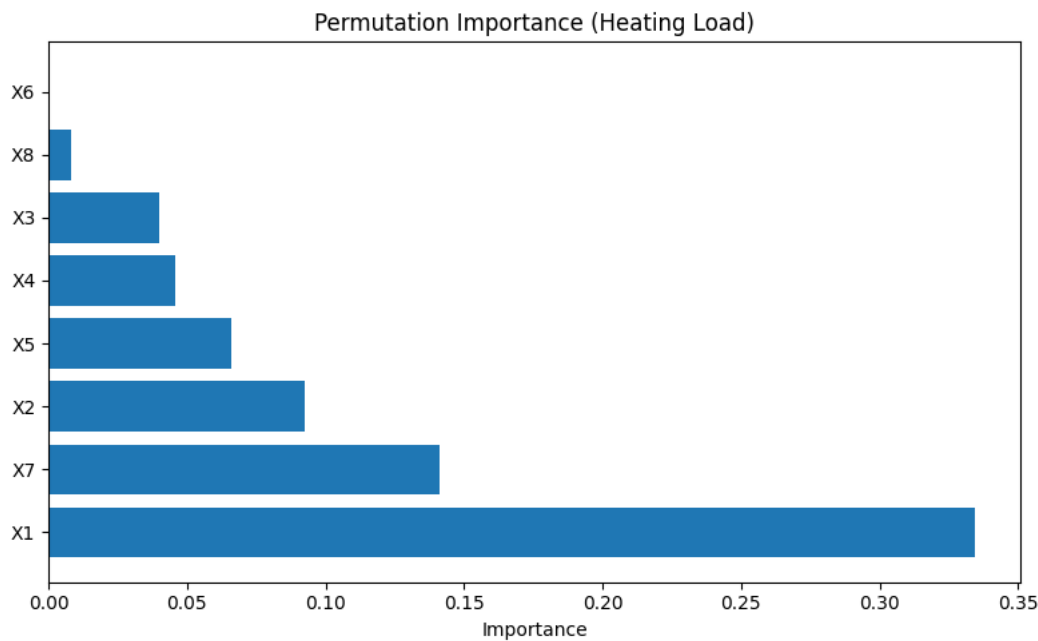


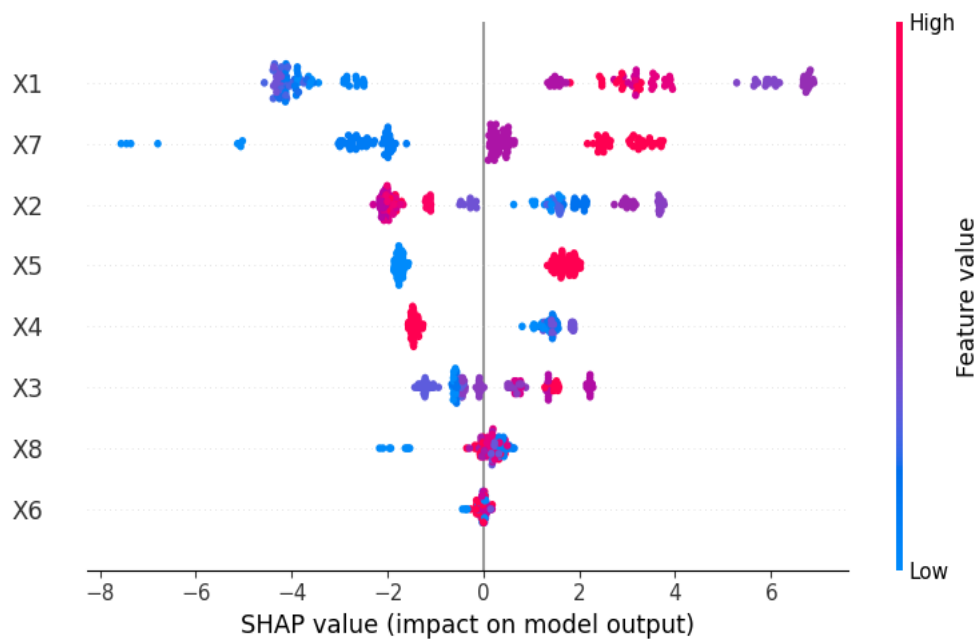
Energy Efficiency Regression Report

Dataset: UCI Energy Efficiency. Target: Heating Load (Y1). Model: RandomForestRegressor.

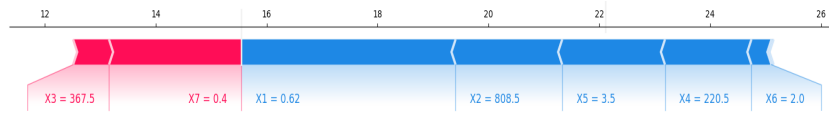
Permutation Importance (PI):



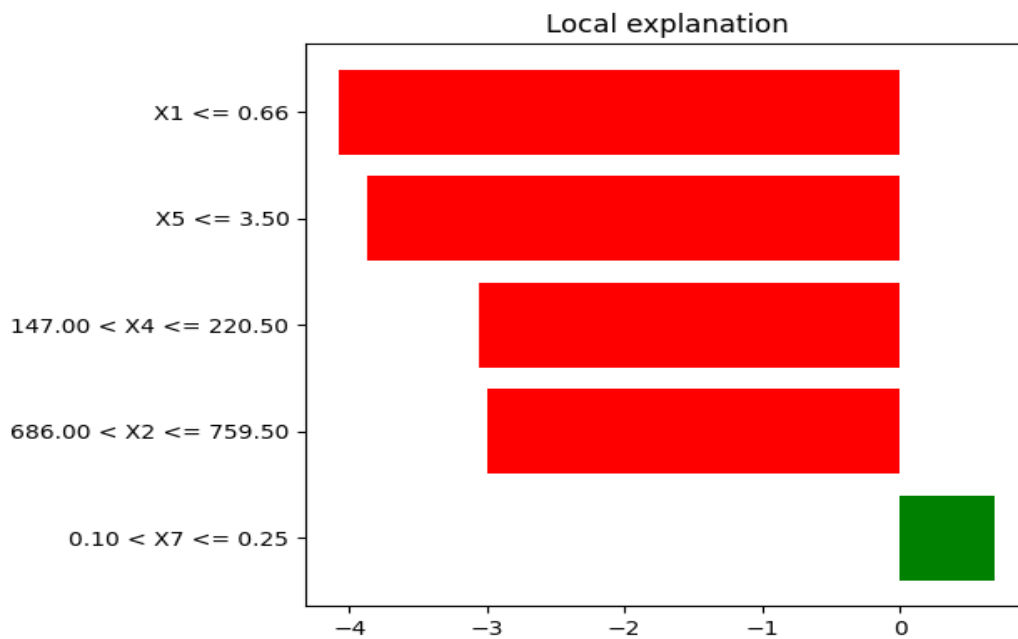
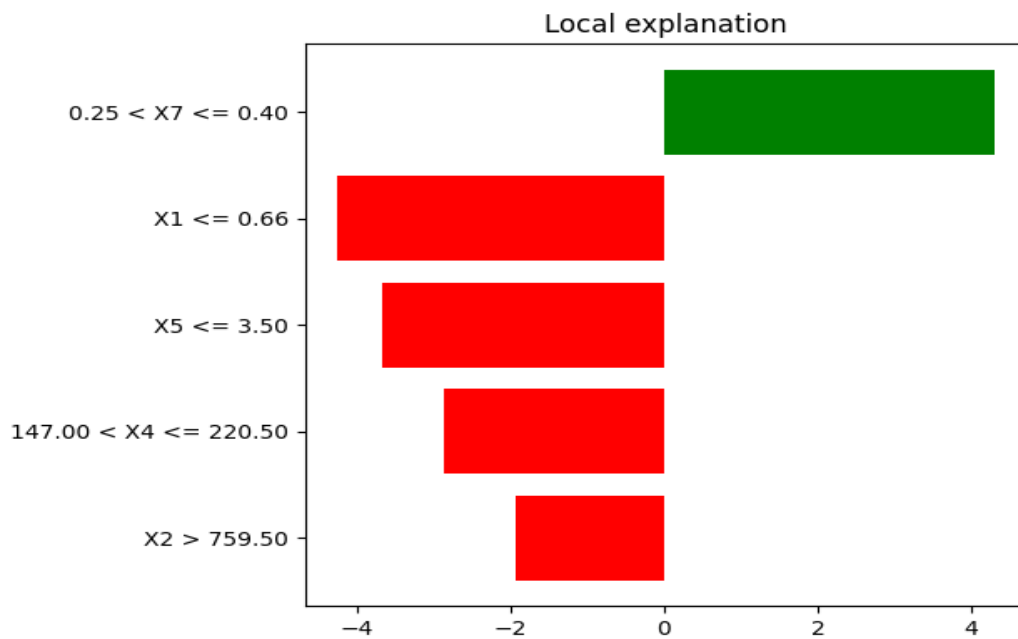
SHAP Global Summary:



SHAP Local Explanation (Force Plot):



LIME Explanations for 2 Buildings:



Insights & Comparison:

Relative compactness, surface area, wall area, and roof area are consistently top predictors. Permutation Importance gives a global ranking but no direction of effect. SHAP provides both global rankings and local positive/negative contributions. LIME highlights feature effects for individual predictions, useful for case studies. All methods agree on the dominance of compactness and surface area features. SHAP and LIME both explain local predictions, but SHAP is more mathematically rigorous.